

Deployment Process and Release Train (2026)

Owner: Infrastructure team · **Approved by:** Priya Natarajan, VP Engineering · **Effective:** 1 September 2026

This document supersedes "Deployment Process (2024 edition)". Where the two documents differ, this version applies. The 2024 edition is kept only for historical reference.

1. Overview

Atlas and Beacon services are deployed to production through a scheduled **release train** that departs twice a week, on **Tuesdays and Thursdays**. Everything merged to the `main` branch before the branch cut is included in the next train. Deployments are executed by the deployment tool `shipit` through GitHub Actions and Argo CD; nobody deploys to production from a laptop.

2. Release train schedule

Step	Time (Lisbon time)	Responsible
Branch cut (<code>release/YYYY-MM-DD</code>)	11:00 on train day	Release captain (automated)
Staging deploy and smoke tests	11:15–13:00	Release captain
Go/no-go check in <code>#release-train</code>	13:30	Release captain and team leads
Production canary starts	14:00	<code>shipit</code>
Full rollout complete	by 16:00	<code>shipit</code>

The release captain rotates weekly among senior engineers and is listed in the `#release-train` channel topic. If a train is cancelled (for example because of an ongoing SEV1 or SEV2), changes roll over to the next train.

3. Canary and rollback

- Every production deploy starts with a **canary at 5% of traffic for 30 minutes** in each region, eu-west-1 first and us-east-2 second.
- `shipit` compares the canary with the baseline. The deploy is **rolled back automatically if the canary error rate is above 2%** or if p99 latency is more than 25% higher than the baseline.
- After a successful canary the rollout proceeds to 25%, 50% and 100% in 15-minute steps.
- Manual rollback: `shipit rollback <service> --to <previous-release>`. A rollback does not need approval and should be the first reaction to a bad deploy.

4. Hotfixes

A hotfix is a change deployed outside the release train. Hotfixes are allowed only to fix a customer-facing defect or a security issue.

- The PR must carry the `hotfix` label and be reviewed within 2 hours (see Code Review Guidelines).
- An Engineering Manager must approve the hotfix in the `#release-train` channel before deployment.
- Hotfixes go through the same canary as regular deploys; the canary duration may be reduced to 10 minutes only for SEV1 mitigations, with Incident Commander approval.
- No hotfixes on Fridays after 15:00 or on weekends, except to mitigate a SEV1 or SEV2.

5. Database migrations

- Migrations must be backwards compatible with the currently deployed version (expand–contract pattern).
- Migrations run as a separate pre-deploy step in the train and must complete in under 10 minutes on production-sized data; longer migrations are executed as background jobs.
- Destructive migrations (dropping columns or tables) need sign-off from the owning team lead and must wait at least two trains after the code no longer uses the column.

6. Feature flags

New user-facing features are shipped dark behind a LaunchDarkly flag. Flags are removed within 60 days of reaching 100% rollout. The Product team decides when a flag is turned on for customers.

7. End-of-year code freeze

To protect customers during the holiday period, a production code freeze applies at the end of every year.

- The **2026 code freeze starts on Friday, 18 December 2026 at 18:00 Lisbon time** and ends on **Monday, 4 January 2027 at 09:00**.
- No release trains run during the freeze. The last train before the freeze departs on Thursday, 17 December 2026.
- During the freeze only hotfixes for SEV1 or SEV2 incidents and security fixes are allowed, each approved by the VP Engineering.
- Infrastructure changes through `infra-live` are also frozen, except for capacity increases.

From 2026 onwards, the freeze always starts on the third Friday of December. (The 2024 edition of this document used a fixed start date of 15 December, which no longer applies.)

8. Responsibilities

Role	Responsibility
Change author	Merges before branch cut, monitors their change during the canary, writes the release note entry
Release captain	Runs the train, makes the go/no-go call, communicates delays
Team lead	Confirms readiness of risky changes at the go/no-go check
Infrastructure team	Owens shipit, Argo CD and the train tooling

9. Release notes

Each customer-visible change requires an entry in the `CHANGELOG.md` of the service. The Product team compiles the public Atlas release notes from these entries for every minor version.

10. Metrics

The Infrastructure team reports monthly on:

- deployment frequency (target: at least 8 trains per month);
- change failure rate (target: below 5% of deploys rolled back);
- mean time to restore after a failed deploy (target: under 30 minutes).

11. Change history

Version	Date	Change
2024 edition	10 September 2024	Original process; freeze fixed to 15 December
2026	25 August 2026	Canary metrics, hotfix rules, freeze moved to third Friday of December (18 December 2026)